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EX NO: 8a

SIMULATION AND ANALYSIS OF BPSK MODULATION USING
CONSTELLATION DIAGRAMS IN MATLAB

Objective

- i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.
- ii) To Plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.
- iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

Matlab code

```
clc;
clear;
close all;

%% Objective 1: Generate BPSK symbols
N = 1000;
data = randi([0 1], N, 1);
% BPSK Mapping
bpsk_symbols = 2*data - 1;
% Number of bits
% Random binary sequence

%% Objective 2: Plot BPSK Signal Constellation
figure;
subplot(1,2,1);
scatter(real(bpsk_symbols), imag(bpsk_symbols), ...
30, 'filled');
```

```

hold on;

xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);

grid on;

axis equal;

xlim([-1.5 1.5]);
ylim([-1 1]);

xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');

%% Objective 3: Add Noise and Evaluate Performance

SNR = 5;

% Signal-to-Noise Ratio (dB)

rx_signal = awgn(bpsk_symbols, SNR, 'measured');

subplot(1,2,2);

scatter(real(rx_signal), imag(rx_signal), ...

15, 'filled');

hold on;

xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);

grid on;

axis equal;

xlim([-2.5 2.5]);
ylim([-2 2]);

xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);

%% Symbol Detection

detected_bits = real(rx_signal) > 0;

```

```

% Calculate Errors

num_errors = sum(data ~= detected_bits);

BER = num_errors/N;

fprintf('Number of Transmitted Bits : %d\n', N);

fprintf('Number of Bit Errors
: %d\n', num_errors);

fprintf('Bit Error Rate (BER) : %f\n', BER);

```

OUTPUT:

